## ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** silver | Silver lies below hydrogen. |
| 2 | **(b)** They have high melting points | Strong forces between ions. |
| 3 | **(b)** keeps air and water away | Barrier effect. |
| 4 | **(b)** sodium | Only highly reactive metals react with cold water. |
| 5 | **(b)** +2 | It loses two electrons. |
| 6 | **(b)** is more reactive than iron | Zinc corrodes first. |
| 7 | **(b)** $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ | Only a more reactive metal displaces a less reactive. |
| 8 | **(b)** $X^-$ | It gains one electron. |
| 9 | **(b)** anodising | Anodising. |
| 10 | **(b)** Mg > Zn > Fe > Cu | Standard reactivity series. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Bubbles of hydrogen stick to its surface (1); They make it lighter than water (1). | 1+1 |
| 12 | Ions are free in solution (1); Fixed in the solid lattice (1). | 1+1 |
| 13 | Water is polar and separates ions (1); Kerosene is non-polar (1). | 1+1 |
| 14 | Copper is below hydrogen in the series (1); Zinc is above hydrogen (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (1); A thin layer of $Al_2O_3$ forms on the surface (1); It prevents further reaction (anodising thickens it) (1). | 1+1+1 |
| 16 | Crushing and grinding (1); Removal of gangue (1); Roasting converts ZnS to ZnO (1). | 1+1+1 |
| 17 | Zinc layer isolates iron from air (1); Zinc is more reactive and gets oxidised first (1); So iron is protected even if scratched (1). | 1+1+1 |
| | **OR:** A > C > B (1); A displaces hydrogen from acid and reacts with cold water (1); B does not react with acids, so it is least reactive (1). | 1+1+1 |
| 18 | Metals from reduction contain impurities (1); Electrolytic refining: impure Cu anode and pure Cu cathode (1); Pure copper deposits on the cathode (1). | 1+1+1 |
| | **OR:** Position in the reactivity series (1); Na, K are at the top (1); Cu is below hydrogen (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Metal oxides are basic (1); Hydrogen is displaced by reactive metals (1). | 1+1+1+1+1 |
| | **OR:** Bridges: painting, galvanisation (1); Aluminium: anodising (1); Ships: sacrificial zinc/magnesium plates (1); Zn, Mg are more reactive than iron (1); They corrode instead of iron (1). | 1+1+1+1+1 |
| 20 | Hg: heating cinnabar, as unreactive (1); $2\,HgS + 3\,O_2 \longrightarrow 2\,HgO + 2\,SO_2$; $2\,HgO \longrightarrow 2\,Hg + O_2$ (1); Zn: calcination and reduction with carbon (1); $ZnO + C \longrightarrow Zn + CO$ (1); Na: electrolysis of molten NaCl (1). | 1+1+1+1+1 |
| | **OR:** Zn displaces Fe and Cu (1); Fe displaces Cu but not Zn (1); Cu displaces none (1); Order Zn > Fe > Cu (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as $Fe_2O_3$; it is $Fe_3O_4$ and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.

- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $\mathrm{FeSO_4}$; a less reactive metal cannot displace a more reactive one.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $MgCl_2$ | $Mg^{2+}$ and $Cl^-$. |
| 2 | **(b)** heating in air | Heating in air promotes oxidation. |
| 3 | **(b)** hydrogen | $HNO_3$ oxidises the hydrogen formed. |
| 4 | **(c)** $Na_2O$ | $Na^+$ and $O^{2-}$. |
| 5 | **(c)** a thin protective layer forms on its surface | A passive protective film forms. |
| 6 | **(d)** mercury | Mercury is unreactive. |
| 7 | **(d)** like charges come close on shifting layers | Repulsion causes breaking. |
| 8 | **(d)** a thin strip of pure copper | $Cu^{2+}$ deposits on it. |
| 9 | **(d)** sulphides and carbonates | Zn, Fe, Pb ores. |
| 10 | **(a)** NaCl | Ionic compounds. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Roasting: sulphide ore in excess air (1); Calcination: carbonate ore in limited air (1). | 1+1 |
| 12 | Iron displaces copper (1); Copper deposits on the nail (1). | 1+1 |
| 13 | Sodium is more reactive than carbon (1); Carbon cannot reduce its oxide; electrolysis is used (1). | 1+1 |
| 14 | Highly reactive, forms stable compounds (1); Only electrolytic reduction can reduce $Na^+$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | X: 2,8,2 loses 2 electrons; Y: 2,8,7 gains 1 (1); $XY_2$ (i.e. $MgCl_2$) (1); Ionic bond (1). | 1+1+1 |
| 16 | Alloys are harder and stronger (1); Adding carbon makes iron hard and brittle (1); Ni and Cr make stainless steel which resists rusting (1). | 1+1+1 |
| 17 | React with acids and bases (1); $Al_2O_3$, ZnO (1); $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1). | 1+1+1 |
| | **OR:** Mg: vigorous bubbles (1); Zn: moderate (1); Cu: no reaction as it is less reactive than hydrogen (1). | 1+1+1 |
| 18 | $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1); $Ca + 2\,H_2O \longrightarrow Ca(OH)_2 + H_2$ (1); Hydrogen is evolved in both (1). | 1+1+1 |
| | **OR:** Copper: basic copper carbonate (1); Silver: silver sulphide (1); Due to reaction with air gases (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | MO is amphoteric (e.g. ZnO) (1); $ZnO + 2\,HCl \longrightarrow ZnCl_2 + H_2O$ (1); $ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (1); M can be zinc (1); Aluminium is another such metal (1). | 1+1+1+1+1 |
| | **OR:** $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1); Al is more reactive than Fe (1); Aluminium reduces $Fe_2O_3$ (1); Highly exothermic; iron is molten (1); Used for welding rails (1). | 1+1+1+1+1 |
| 20 | Na: violent and exothermic (1); Ca: less violent, floats (1); Mg: reacts with hot water (1); Al, Fe: only with steam (1); Difference arises from reactivity (1). | 1+1+1+1+1 |
| | **OR:** Ionic: high melting point; NaCl (1); Covalent: low; $CH_4$ (1); Ionic conduct in solution (1); Covalent do not (1); Ionic soluble in water; covalent in organic solvents (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as $Fe_2O_3$; it is $Fe_3O_4$ and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.
- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $FeSO_4$; a less reactive metal cannot displace a more reactive one.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** silver and gold | Less reactive metals do not dissolve. |
| 2 | **(b)** aluminium | Al is more reactive than Fe. |
| 3 | **(a)** carbon | Non-metals share electrons. |
| 4 | **(c)** liquid and expands uniformly | Property of mercury. |
| 5 | **(a)** sodium | $Na^+ + e^- \longrightarrow Na$. |
| 6 | **(c)** 3 | Al (2,8,3) loses 3. |
| 7 | **(c)** solder | Low melting point. |
| 8 | **(b)** aluminium | $Al_2O_3$ layer. |
| 9 | **(a)** $AlCl_3$ | $Al^{3+}$ and $Cl^-$. |
| 10 | **(b)** sodium chloride | It is a compound. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Welding railway tracks (1); $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ (1). | 1+1 |
| 12 | Black coating of copper(II) oxide (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1). | 1+1 |
| 13 | $ZnO + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2O$ (2). | 2 |
| 14 | 24-carat gold is pure gold (1); 22-carat is alloyed with copper/silver to be harder (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Aqua regia is 3 parts HCl to 1 part $HNO_3$ (1); $HNO_3$ oxidises gold; chloride forms soluble complex (1); Gold dissolves slowly (1). | 1+1+1 |
| 16 | Sodium chloride (1); Electron transferred from Na to Cl (1); Electrostatic force makes an ionic bond (1). | 1+1+1 |
| 17 | Flux combines with gangue (1); Slag is formed (1); Slag is removed (1). | 1+1+1 |
| | **OR:** $2\,Cu_2S + 3\,O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2$ (1); $2\,Cu_2O + Cu_2S \longrightarrow 6\,Cu + SO_2$ (1); Roasting then auto-reduction (1). | 1+1+1 |
| 18 | Metals have free electrons (1); Graphite layers have delocalised electrons (1); Diamond has none (1). | 1+1+1 |
| | **OR:** (i) Iron displaces copper (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); (ii) No reaction, since zinc is more reactive (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Mg loses 2, O gains 2 (1); Na loses 1; O gains 2, so $Na_2O$ (1); High melting points (1); Brittle (1); Conduct in molten state (1). | 1+1+1+1+1 |
| | **OR:** Alloys are harder and more resistant (1); Gold is soft (1); 22-carat has Ag/Cu added (1); Brass (1); Bronze (1). | 1+1+1+1+1 |
| 20 | Anode: impure copper (1); Cathode: pure copper (1); Electrolyte acidified $CuSO_4$ (1); Copper dissolves from anode and deposits on cathode (1); Impurities settle as anode mud (1). | 1+1+1+1+1 |
| | **OR:** $Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O$ (1); $Al_2O_3 + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2O$ (1); Zn reacts with acid and base (1); $Zn + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2$ (1); Oxide layer protects metal (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Storing sodium or potassium in water; they are stored under kerosene because they react violently with water and air.
- Confusing **roasting** (sulphide ore, excess air) with **calcination** (carbonate ore, limited air).
- Writing the product of iron with steam as $Fe_2O_3$; it is $Fe_3O_4$ and hydrogen gas is released.
- Thinking that all metals are hard solids and all non-metals are non-conductors; Na is soft, Hg is a liquid and graphite conducts.
- Saying that an ionic solid conducts electricity; it conducts only when molten or in aqueous solution.
- Writing that copper can displace hydrogen from dilute acids or iron from $FeSO_4$; a less reactive metal cannot displace a more reactive one.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND